

sub-modules: `usb.core` as the main module and `usb.util` containing utility functions.

PyUSB 1.0 tutorial gives the basic information about the complex USB protocol. If you want to know all information related to USB, refer the books *USB Design by Example* by John Hyde and *USB Complete* by Jan Axelson.

Functions needed for USB configuration and data acquisition are incorporated in `pic18f4550.py`, namely, `__init__(self, val1, val2)`, `set_portb_output(self, val)`, `set_portd_output(self, val)`, `read_analog_0(self)`, `read_analog_1(self)` and `set_voltage12(self, val)`.

Details are as follows:

- `__init__(self, val1, val2)` takes care of USB communication.
- `set_portb_output(self, val)` helps set port B values from 0 to 255.
- `set_portd_output(self, val)` helps set port D values from 0 to 255.
- `read_analog_0(self)` helps read analogue values from channel 0.

- `read_analog_1(self)` helps read analogue values from channel 1.
- `set_voltage12(self, val)` helps send a 12-bit value to DAC MAX5154.

**Software installation.** Install Python2.5.2 IDE or higher in Windows 7 operating system. Then, install the supporting tools one by one. Default folder is C folder (C:\Python25). Extract PyUSB-1.0.0a2 zip file to Python25 folder. Now, using Command Prompt, go to PyUSB folder, which is inside Python25 folder, and install PyUSB using the following command (also shown in Fig. 1):

```
>python setup.py install
```

Having installed the necessary software, place `pic18f4550.py` and

`efy_trans.py` files inside Python25 folder.

## Circuit and working

Circuit diagram of the USB interface using Python and transistor curve tracer is shown in Fig. 2. It is built around PIC18F4550 microcontroller, MAX5154 12-bit digital-to-analogue converter, voltage regulator AD780, voltage amplifier LM358, current-to-voltage converter OP741, power transistor SL100/CL100, transistor under study BC547 (T1) and other essential components needed for USB configuration.

The microcontroller has an on-chip USB transceiver, which is connected to the host PC through a USB cable. Clock

frequency required for full-speed USB operation is derived from external 20MHz crystal. A pull-up resistor internal to the microcontroller configures the USB device as a full-speed device. In this design, power required for PIC18F4550 operation is drawn from the bus.

Base current of the transistor under study is varied from 0µA to 175µA in eight steps using port

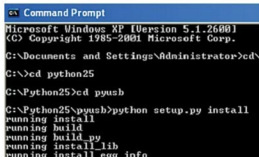


Fig. 1: Command Prompt screenshot

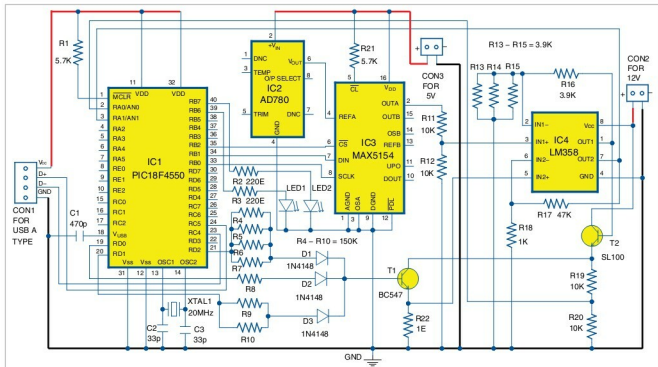


Fig. 2: Circuit diagram of the USB interface and transistor curve tracer